

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : PYTHON FOR DATA SCIENCE

Practical No : 9

Aim 9A: Create and Explore a Dataset Using Pandas

Create a student dataset containing Student ID, Name, Age, Gender, Course, and Marks. Store the data in a Pandas DataFrame and perform the following:

1. Display the complete dataset.
2. Display the first 5 records.
3. Display the last 5 records.
4. Display the number of rows and columns.
5. Display column names.
6. Display basic information about the dataset.
7. Display statistical information about numerical columns.

CODE :

```
import pandas as pd

print("Om Wala S119")

data = {

    "Student_ID": [101, 102, 103, 104, 105, 106, 107, 108],

    "Name": ["Om", "Shreyash", "Krish", "Deepak", "Palak", "Sahil", "Vedant",
"Krishna"],

    "Age": [19, 20, 19, 21, 20, 19, 21, 20],

    "Gender": ["Male", "Male", "Male", "Female",
    "Female", "Male", "Male", "Male"],

    "Course": ["BSc CS", "BSc CS", "BSc IT", "BSc CS",
    "BSc IT", "BSc CS", "BSc IT", "BSc CS"],

    "Marks": [98, 93, 67, 92, 88, 74, 81, 69]

}

df = pd.DataFrame(data)

print("Complete Dataset:")
```

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```
print(df)

print("\nFirst 5 Records:")

print(df.head())

print("\nLast 5 Records:")

print(df.tail())

print("\nShape:")

print(df.shape)

print("\nColumn Names:")

print(df.columns)

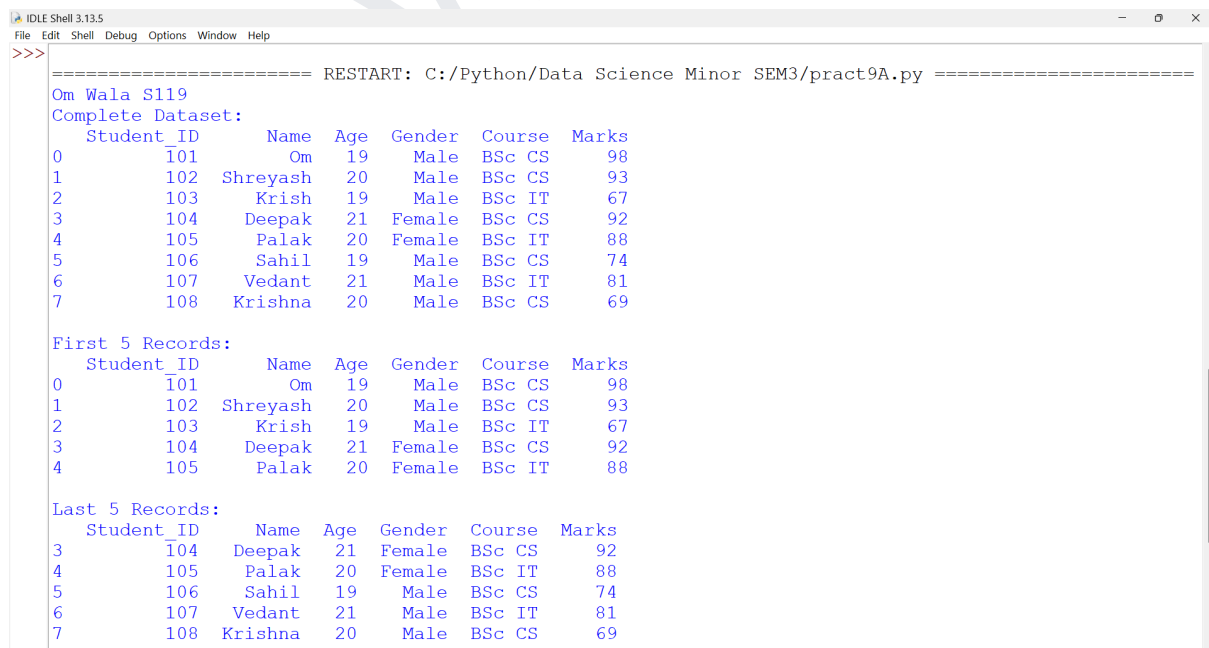
print("\nDataset Information:")

df.info()

print("\nStatistical Information:")

print(df.describe())
```

OUTPUT :



```
>>>===== RESTART: C:/Python/Data Science Minor SEM3/pract9A.py =====
Om Wala S119
Complete Dataset:
  Student_ID  Name  Age  Gender  Course  Marks
0         101    Om   19    Male  BSc CS    98
1         102 Shreyash  20    Male  BSc CS    93
2         103   Krish  19    Male  BSc IT    67
3         104  Deepak  21  Female  BSc CS    92
4         105   Palak  20  Female  BSc IT    88
5         106   Sahil  19    Male  BSc CS    74
6         107  Vedant  21    Male  BSc IT    81
7         108  Krishna  20    Male  BSc CS    69

First 5 Records:
  Student_ID  Name  Age  Gender  Course  Marks
0         101    Om   19    Male  BSc CS    98
1         102 Shreyash  20    Male  BSc CS    93
2         103   Krish  19    Male  BSc IT    67
3         104  Deepak  21  Female  BSc CS    92
4         105   Palak  20  Female  BSc IT    88

Last 5 Records:
  Student_ID  Name  Age  Gender  Course  Marks
3         104  Deepak  21  Female  BSc CS    92
4         105   Palak  20  Female  BSc IT    88
5         106   Sahil  19    Male  BSc CS    74
6         107  Vedant  21    Male  BSc IT    81
7         108  Krishna  20    Male  BSc CS    69
```

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```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help

Shape:
(8, 6)

Column Names:
Index(['Student_ID', 'Name', 'Age', 'Gender', 'Course', 'Marks'], dtype='str')

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Student_ID  8 non-null      int64
1   Name        8 non-null      str
2   Age         8 non-null      int64
3   Gender      8 non-null      str
4   Course      8 non-null      str
5   Marks       8 non-null      int64
dtypes: int64(3), str(3)
memory usage: 516.0 bytes

Statistical Information:
      Student_ID      Age      Marks
count      8.000000      8.000000      8.000000
mean      104.50000      19.875000      82.750000
std         2.44949      0.834523      11.756457
min       101.00000      19.000000      67.000000
25%       102.75000      19.000000      72.750000
50%       104.50000      20.000000      84.500000
75%       106.25000      20.250000      92.250000
max       108.00000      21.000000      98.000000
```

Aim 9B : Read and Explore a CSV Dataset

Create a CSV file named students.csv containing student information:

- Student_ID
- Name
- Gender
- Course
- Marks
- Attendance

Write a Python program to:

1. Read the CSV file using Pandas.
2. Display the dataset.
3. Display the first 10 records.
4. Find the number of students.
5. Display the column names.
6. Find the average marks.
7. Find the average attendance.

CODE :

```
import pandas as pd
```

```
df = pd.read_csv("movies.csv")
```

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```
print("Movie Dataset:")

print(df)

print("\nFirst 10 Movie Records:")

print(df.head(10))

print("\nNumber of Movies:")

print(len(df))

print("\nColumn Names:")

print(df.columns)

print("\nAverage Movie Rating:")

print(df["Rating"].mean())

print("\nAverage Movie Duration:")

print(df["Duration"].mean())

print("\nOm Wala S119")
```

OUTPUT :

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Python/Data Science Minor SEM3/pract9B.py =====
Movie Dataset:
  Movie_ID      Title  ... Rating  Duration
0         1      Inception  ...    8.8      148
1         2  The Dark Knight  ...    9.0      152
2         3   Interstellar  ...    8.7      169
3         4        Avatar  ...    7.8      162
4         5       Titanic  ...    7.9      195
5         6  The Shawshank Redemption  ...    9.3      142
6         7   Forrest Gump  ...    8.8      142
7         8    The Matrix  ...    8.7      136
8         9    Gladiator  ...    8.5      155
9        10     Parasite  ...    8.5      132
10       11        Joker  ...    8.3      122
11       12  Avengers Endgame  ...    8.4      181
12       13   The Godfather  ...    9.2      175
13       14   The Lion King  ...    8.5       88
14       15     3 Idiots  ...    8.4      170

[15 rows x 7 columns]
```

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```
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File Edit Shell Debug Options Window Help

First 10 Movie Records:
  Movie_ID      Title      Genre  ... Release_Year  Rating  Duration
0         1      Inception  Sci-Fi  ...      2010      8.8      148
1         2    The Dark Knight  Action  ...      2008      9.0      152
2         3    Interstellar  Sci-Fi  ...      2014      8.7      169
3         4        Avatar  Sci-Fi  ...      2009      7.8      162
4         5        Titanic  Romance  ...      1997      7.9      195
5         6  The Shawshank Redemption  Drama  ...      1994      9.3      142
6         7    Forrest Gump  Drama  ...      1994      8.8      142
7         8    The Matrix  Sci-Fi  ...      1999      8.7      136
8         9    Gladiator  Action  ...      2000      8.5      155
9        10      Parasite  Thriller  ...      2019      8.5      132

[10 rows x 7 columns]

Number of Movies:
15

Column Names:
Index(['Movie_ID', 'Title', 'Genre', 'Director', 'Release_Year', 'Rating',
      'Duration'],
      dtype='str')

Average Movie Rating:
8.586666666666668

Average Movie Duration:
151.26666666666668

Om Wala S119
>>>
```

Aim 9C : Selecting and Filtering Data

Using the students.csv dataset, perform the following:

1. Display only Name and Marks.
2. Display students who scored more than 75 marks.
3. Display students whose attendance is more than 80%.
4. Display female students.
5. Display students from the BSc CS course.
6. Display students who scored more than 75 marks and have attendance above 80%.

CODE :

```
import pandas as pd
```

```
df = pd.read_csv("movies.csv")
```

```
print("Title and Rating:")
print(df[["Title", "Rating"]])
```

```
print("\nMovies with rating greater than 8.0:")
print(df[df["Rating"] > 8.0])
```

```
print("\nMovies released after 2010:")
print(df[df["Release_Year"] > 2010])
```

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```
print("\nSci-Fi Movies:")
print(df[df["Genre"] == "Sci-Fi"])

print("\nMovies directed by Christopher Nolan:")
print(df[df["Director"] == "Christopher Nolan"])

print("\nMovies with rating greater than 8.0 and duration less than 160 minutes:")
print(df[(df["Rating"] > 8.0) & (df["Duration"] < 160)])

print("\nOm Wala S119")
```

OUTPUT :

```
===== RESTART: C:/Python/Data Science Minor SEM3/pract9C.py =====
Title and Rating:
   Title  Rating
0  Inception    8.8
1  The Dark Knight    9.0
2  Interstellar    8.7
3  Avatar    7.8
4  Titanic    7.9
5  The Shawshank Redemption    9.3
6  Forrest Gump    8.8
7  The Matrix    8.7
8  Gladiator    8.5
9  Parasite    8.5
10 Joker    8.3
11 Avengers Endgame    8.4
12 The Godfather    9.2
13 The Lion King    8.5
14 3 Idiots    8.4

Movies with rating greater than 8.0:
   Movie_ID  Title  ... Rating Duration
0         1  Inception  ...    8.8    148
1         2  The Dark Knight  ...    9.0    152
2         3  Interstellar  ...    8.7    169
5         6  The Shawshank Redemption  ...    9.3    142
6         7  Forrest Gump  ...    8.8    142
7         8  The Matrix  ...    8.7    136
8         9  Gladiator  ...    8.5    155
9        10  Parasite  ...    8.5    132
10       11  Joker  ...    8.3    122
11       12  Avengers Endgame  ...    8.4    181
12       13  The Godfather  ...    9.2    175
13       14  The Lion King  ...    8.5     88
14       15  3 Idiots  ...    8.4    170

[13 rows x 7 columns]

Movies released after 2010:
   Movie_ID  Title  Genre  ... Release_Year  Rating  Duration
2         3  Interstellar  Sci-Fi  ...    2014    8.7    169
9        10  Parasite  Thriller  ...    2019    8.5    132
10       11  Joker  Drama  ...    2019    8.3    122
11       12  Avengers Endgame  Action  ...    2019    8.4    181

[4 rows x 7 columns]
```

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```
Sci-Fi Movies:
  Movie_ID      Title  Genre  ... Release_Year  Rating  Duration
0         1    Inception  Sci-Fi  ...      2010      8.8      148
2         3  Interstellar  Sci-Fi  ...      2014      8.7      169
3         4      Avatar  Sci-Fi  ...      2009      7.8      162
7         8   The Matrix  Sci-Fi  ...      1999      8.7      136

[4 rows x 7 columns]

Movies directed by Christopher Nolan:
  Movie_ID      Title  Genre  ... Release_Year  Rating  Duration
0         1    Inception  Sci-Fi  ...      2010      8.8      148
1         2  The Dark Knight  Action  ...      2008      9.0      152
2         3    Interstellar  Sci-Fi  ...      2014      8.7      169

[3 rows x 7 columns]

Movies with rating greater than 8.0 and duration less than 160 minutes:
  Movie_ID      Title  ... Rating  Duration
0         1    Inception  ...      8.8      148
1         2  The Dark Knight  ...      9.0      152
5         6  The Shawshank Redemption  ...      9.3      142
6         7    Forrest Gump  ...      8.8      142
7         8    The Matrix  ...      8.7      136
8         9    Gladiator  ...      8.5      155
9        10    Parasite  ...      8.5      132
10       11      Joker  ...      8.3      122
13       14  The Lion King  ...      8.5       88

[9 rows x 7 columns]

Om Wala S119
>>>
```

Aim 9D : Sorting and Ranking Dataset

Question:

Using the student dataset:

1. Sort students according to marks in ascending order.
2. Sort students according to marks in descending order.
3. Sort students according to attendance in descending order.
4. Display the top 5 students based on marks.
5. Display the bottom 3 students based on marks.

CODE :

```
import pandas as pd
```

```
df = pd.read_csv("movies.csv")
```

```
print("Rating in Ascending Order:")
```

```
print(df.sort_values("Rating"))
```

```
print("\nRating in Descending Order:")
```

```
print(df.sort_values("Rating", ascending=False))
```

```
print("\nDuration in Descending Order:")
```

```
print(df.sort_values("Duration", ascending=False))
```

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```
print("\nTop 5 Movies:")  
print(df.sort_values("Rating", ascending=False).head(5))
```

```
print("\nBottom 3 Movies:")  
print(df.sort_values("Rating").head(3))
```

```
print("\nOm Wala S119")
```

OUTPUT :

```
===== RESTART: C:/Python/Data Science Minor SEM3/pract9D.py =====  
Rating in Ascending Order:  
   Movie_ID  Title  ... Rating Duration  
3         4  Avatar  ...    7.8     162  
4         5  Titanic  ...    7.9     195  
10        11    Joker  ...    8.3     122  
11        12  Avengers Endgame  ...    8.4     181  
14        15    3 Idiots  ...    8.4     170  
9         10  Parasite  ...    8.5     132  
13        14  The Lion King  ...    8.5      88  
8         9   Gladiator  ...    8.5     155  
2         3  Interstellar  ...    8.7     169  
7         8  The Matrix  ...    8.7     136  
0         1   Inception  ...    8.8     148  
6         7  Forrest Gump  ...    8.8     142  
1         2  The Dark Knight  ...    9.0     152  
12        13  The Godfather  ...    9.2     175  
5         6  The Shawshank Redemption  ...    9.3     142  
  
[15 rows x 7 columns]  
  
Rating in Descending Order:  
   Movie_ID  Title  ... Rating Duration  
5         6  The Shawshank Redemption  ...    9.3     142  
12        13  The Godfather  ...    9.2     175  
1         2  The Dark Knight  ...    9.0     152  
6         7  Forrest Gump  ...    8.8     142  
0         1   Inception  ...    8.8     148  
7         8  The Matrix  ...    8.7     136  
2         3  Interstellar  ...    8.7     169  
13        14  The Lion King  ...    8.5      88  
8         9   Gladiator  ...    8.5     155  
9         10  Parasite  ...    8.5     132  
11        12  Avengers Endgame  ...    8.4     181  
14        15    3 Idiots  ...    8.4     170  
10        11    Joker  ...    8.3     122  
4         5  Titanic  ...    7.9     195  
3         4  Avatar  ...    7.8     162  
  
[15 rows x 7 columns]
```

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```
Duration in Descending Order:
  Movie_ID      Title  ... Rating Duration
4         5      Titanic  ...   7.9     195
11        12  Avengers Endgame  ...   8.4     181
12        13    The Godfather  ...   9.2     175
14        15      3 Idiots  ...   8.4     170
2         3    Interstellar  ...   8.7     169
3         4      Avatar  ...   7.8     162
8         9    Gladiator  ...   8.5     155
1         2    The Dark Knight  ...   9.0     152
0         1      Inception  ...   8.8     148
6         7    Forrest Gump  ...   8.8     142
5         6  The Shawshank Redemption  ...   9.3     142
7         8    The Matrix  ...   8.7     136
9        10      Parasite  ...   8.5     132
10       11      Joker  ...   8.3     122
13       14    The Lion King  ...   8.5      88

[15 rows x 7 columns]

Top 5 Movies:
  Movie_ID      Title  Genre  ... Release_Year  Rating  Duration
5         6  The Shawshank Redemption  Drama  ...   1994     9.3     142
12        13    The Godfather  Crime  ...   1972     9.2     175
1         2    The Dark Knight  Action  ...   2008     9.0     152
6         7    Forrest Gump  Drama  ...   1994     8.8     142
0         1      Inception  Sci-Fi  ...   2010     8.8     148

[5 rows x 7 columns]

Bottom 3 Movies:
  Movie_ID      Title  Genre  Director  Release_Year  Rating  Duration
3         4      Avatar  Sci-Fi  James Cameron  2009     7.8     162
4         5      Titanic  Romance  James Cameron  1997     7.9     195
10        11      Joker  Drama  Todd Phillips  2019     8.3     122

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>>>
```

Aim 9E : Handling Missing Values

Consider a student dataset containing missing values in the Marks and Attendance columns.

Perform the following:

1. Identify missing values.
2. Count missing values in each column.
3. Replace missing marks with the average marks.
4. Replace missing attendance with the average attendance.
5. Display the cleaned dataset.

CODE :

```
import pandas as pd
```

```
df = pd.read_csv("movies.csv")
```

```
print("Original Movie Dataset:")
```

```
print(df)
```

```
print("\nMissing Values:")
```

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```
print(df.isnull())

print("\nCount of Missing Values:")
print(df.isnull().sum())

df["Rating"] = df["Rating"].fillna(df["Rating"].mean())

df["Duration"] = df["Duration"].fillna(df["Duration"].mean())

print("\nCleaned Movie Dataset:")
print(df)

print("\nOm Wala S119")
```

OUTPUT :

```
===== RESTART: C:/Python/Data Science Minor SEM3/pract9E.py =====
Original Movie Dataset:
   Movie_ID  Title  ... Rating Duration
0         1  Inception  ...    8.8   148.0
1         2  The Dark Knight  ...    9.0   152.0
2         3  Interstellar  ...   NaN   169.0
3         4  Avatar  ...    7.8   162.0
4         5  Titanic  ...    7.9    NaN
5         6  The Shawshank Redemption  ...    9.3   142.0
6         7  Forrest Gump  ...    8.8   142.0
7         8  The Matrix  ...   NaN   136.0
8         9  Gladiator  ...    8.5   155.0
9        10  Parasite  ...    8.5    NaN
10       11  Joker  ...    8.3   122.0
11       12  Avengers Endgame  ...    8.4   181.0
12       13  The Godfather  ...    9.2   175.0
13       14  The Lion King  ...    8.5    88.0
14       15  3 Idiots  ...    8.4   170.0

[15 rows x 7 columns]

Missing Values:
   Movie_ID  Title  Genre  Director  Release_Year  Rating  Duration
0     False  False  False    False         False    False    False
1     False  False  False    False         False    False    False
2     False  False  False    False         False    True     False
3     False  False  False    False         False    False    False
4     False  False  False    False         False    False    True
5     False  False  False    False         False    False    False
6     False  False  False    False         False    False    False
7     False  False  False    False         False    True     False
8     False  False  False    False         False    False    False
9     False  False  False    False         False    False    True
10    False  False  False    False         False    False    False
11    False  False  False    False         False    False    False
12    False  False  False    False         False    False    False
13    False  False  False    False         False    False    False
14    False  False  False    False         False    False    False

Count of Missing Values:
Movie_ID      0
Title         0
Genre         0
Director      0
Release_Year  0
Rating        2
Duration      2
dtype: int64
```

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```
Cleaned Movie Dataset:
  Movie_ID      Title  ...  Rating  Duration
0         1    Inception  ...  8.800000  148.000000
1         2  The Dark Knight  ...  9.000000  152.000000
2         3    Interstellar  ...  8.569231  169.000000
3         4      Avatar  ...  7.800000  162.000000
4         5     Titanic  ...  7.900000  149.384615
5         6  The Shawshank Redemption  ...  9.300000  142.000000
6         7    Forrest Gump  ...  8.800000  142.000000
7         8    The Matrix  ...  8.569231  136.000000
8         9     Gladiator  ...  8.500000  155.000000
9        10     Parasite  ...  8.500000  149.384615
10       11      Joker  ...  8.300000  122.000000
11       12  Avengers Endgame  ...  8.400000  181.000000
12       13    The Godfather  ...  9.200000  175.000000
13       14    The Lion King  ...  8.500000   88.000000
14       15      3 Idiots  ...  8.400000  170.000000

[15 rows x 7 columns]

Om Wala S119
>>>
```

Aim 9F : Data Cleaning and Adding New Columns

Using the student dataset, create a new column called Result.

The student should be:

- "Pass" if Marks ≥ 40
- "Fail" if Marks < 40

Also create a Grade column using:

Marks	Grade
90–100	A+
80–89	A
70–79	B
60–69	C
50–59	D
Below 50	F

CODE :

```
import pandas as pd
```

```
df = pd.read_csv("movies.csv")
```

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```
df["Rating_Result"] = df["Rating"].apply(
    lambda x: "Highly Rated" if x >= 8.5 else "Average Rated"
)
```

```
def calculate_category(rating):
    if rating >= 9.0:
        return "Excellent"
    elif rating >= 8.0:
        return "Very Good"
    elif rating >= 7.0:
        return "Good"
    else:
        return "Average"
```

```
df["Rating_Category"] = df["Rating"].apply(calculate_category)
```

```
print(df)
```

```
print("\nOm Wala S119")
```

OUTPUT :

```
===== RESTART: C:/Python/Data Science Minor SEM3/pract9F.py =====
   Movie_ID  Title  ... Rating_Result Rating_Category
0         1  Inception  ...   Highly Rated   Very Good
1         2  The Dark Knight  ...   Highly Rated   Excellent
2         3  Interstellar  ...   Highly Rated   Very Good
3         4  Avatar  ... Average Rated   Good
4         5  Titanic  ... Average Rated   Good
5         6  The Shawshank Redemption  ...   Highly Rated   Excellent
6         7  Forrest Gump  ...   Highly Rated   Very Good
7         8  The Matrix  ...   Highly Rated   Very Good
8         9  Gladiator  ...   Highly Rated   Very Good
9        10  Parasite  ...   Highly Rated   Very Good
10       11  Joker  ... Average Rated   Very Good
11       12  Avengers Endgame  ... Average Rated   Very Good
12       13  The Godfather  ...   Highly Rated   Excellent
13       14  The Lion King  ...   Highly Rated   Very Good
14       15  3 Idiots  ... Average Rated   Very Good

[15 rows x 9 columns]

Om Wala S119
>>>
```

Aim 9G : Statistical Analysis of Dataset

Using students.csv, calculate:

1. Average marks.
2. Maximum marks.
3. Minimum marks.
4. Median marks.
5. Standard deviation of marks.
6. Average attendance.

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7. Number of students.
8. Number of students who scored more than 75.

CODE :

```
import pandas as pd

df = pd.read_csv("movies.csv")

print("Average Rating:", df["Rating"].mean())

print("Maximum Rating:", df["Rating"].max())

print("Minimum Rating:", df["Rating"].min())

print("Median Rating:", df["Rating"].median())

print("Standard Deviation:", df["Rating"].std())

print("Average Duration:", df["Duration"].mean())

print("Number of Movies:", df["Title"].count())

print("Movies with rating above 8.0:",
      (df["Rating"] > 8.0).sum())

print("\nOm Wala S119")
```

OUTPUT :

```
===== RESTART: C:/Python/Data Science Minor SEM3/pract9G.py =====
Average Rating: 8.586666666666668
Maximum Rating: 9.3
Minimum Rating: 7.8
Median Rating: 8.5
Standard Deviation: 0.4172472146045293
Average Duration: 151.26666666666668
Number of Movies: 15
Movies with rating above 8.0: 13

Om Wala S119
>>>
```

Aim 9H : Grouping and Aggregation

Question:

Using the student dataset, perform analysis based on the **Course** column:

1. Find the number of students in each course.
2. Find the average marks for each course.
3. Find the maximum marks for each course.
4. Find the average attendance for each course.

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CODE :

```
import pandas as pd

df = pd.read_csv("movies.csv")

print("Number of Movies in Each Genre:")
print(df.groupby("Genre")["Title"].count())

print("\nAverage Rating by Genre:")
print(df.groupby("Genre")["Rating"].mean())

print("\nMaximum Rating by Genre:")
print(df.groupby("Genre")["Rating"].max())

print("\nAverage Duration by Genre:")
print(df.groupby("Genre")["Duration"].mean())

print("\nOm Wala S119")
```

OUTPUT :

```
>>> ===== RESTART: C:/Python/Data Science Minor SEM3/pract9H.py =====
Number of Movies in Each Genre:
Genre
Action      3
Animation    1
Comedy       1
Crime        1
Drama        3
Romance      1
Sci-Fi       4
Thriller     1
Name: Title, dtype: int64

Average Rating by Genre:
Genre
Action      8.633333
Animation    8.500000
Comedy       8.400000
Crime        9.200000
Drama        8.800000
Romance      7.900000
Sci-Fi       8.500000
Thriller     8.500000
Name: Rating, dtype: float64
```

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```
Maximum Rating by Genre:
Genre
Action      9.0
Animation    8.5
Comedy       8.4
Crime        9.2
Drama        9.3
Romance      7.9
Sci-Fi       8.8
Thriller     8.5
Name: Rating, dtype: float64

Average Duration by Genre:
Genre
Action      162.666667
Animation    88.000000
Comedy       170.000000
Crime        175.000000
Drama        135.333333
Romance      195.000000
Sci-Fi       153.750000
Thriller     132.000000
Name: Duration, dtype: float64

Om Wala S119
>>>
```

Aim 9I : Data Visualization Using Matplotlib

Using the student dataset, create the following charts:

1. Bar chart showing student names and marks.
2. Line chart showing student names and marks.
3. Histogram showing the distribution of marks.
4. Pie chart showing the number of students in each course.

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("movies.csv")
```

```
plt.figure(figsize=(10, 5))
plt.bar(df["Title"], df["Rating"])
plt.xlabel("Movie Title")
plt.ylabel("Rating")
plt.title("Movie Ratings")
plt.xticks(rotation=45)
```

```
plt.figure(figsize=(10, 5))
plt.plot(df["Title"], df["Rating"], marker="o")
plt.xlabel("Movie Title")
plt.ylabel("Rating")
plt.title("Movie Ratings - Line Chart")
plt.xticks(rotation=45)
```

```
plt.figure(figsize=(8, 5))
```

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```
plt.hist(df["Rating"], bins=5)
plt.xlabel("Rating")
plt.ylabel("Number of Movies")
plt.title("Distribution of Movie Ratings")

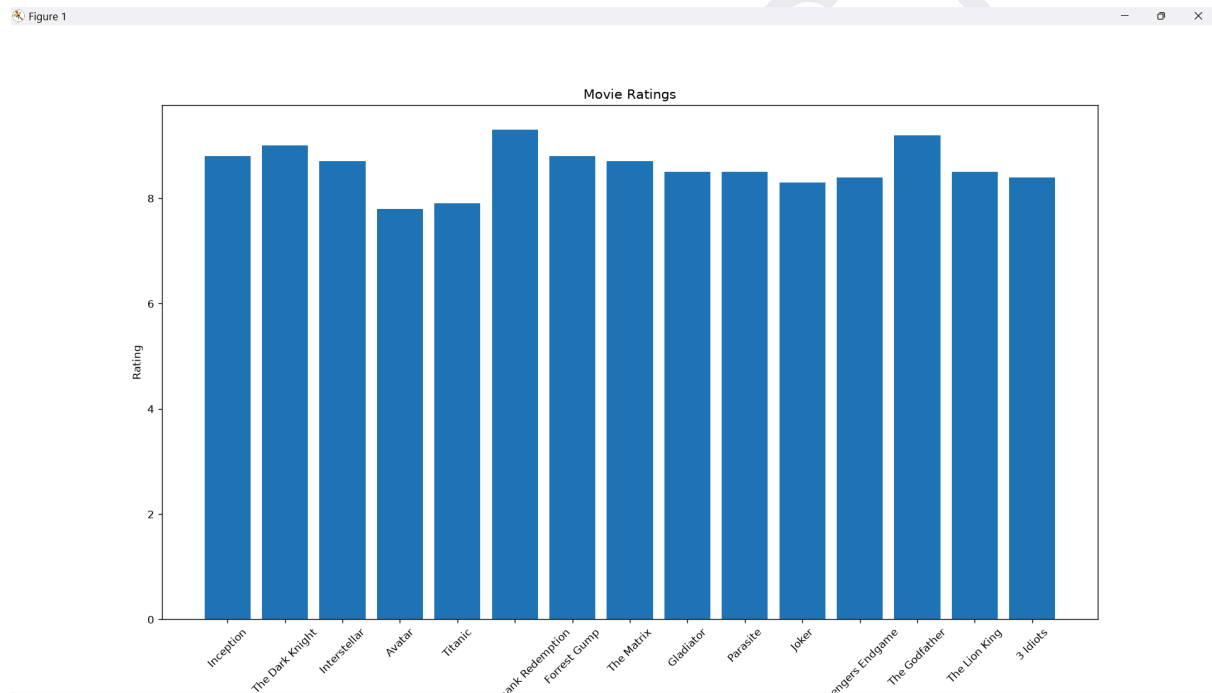
genre_count = df["Genre"].value_counts()

plt.figure(figsize=(7, 7))
plt.pie(genre_count, labels=genre_count.index,
        autopct="%1.1f%%")
plt.title("Movies by Genre")

plt.show()

print("Om Wala S119")
```

OUTPUT :



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Figure 2

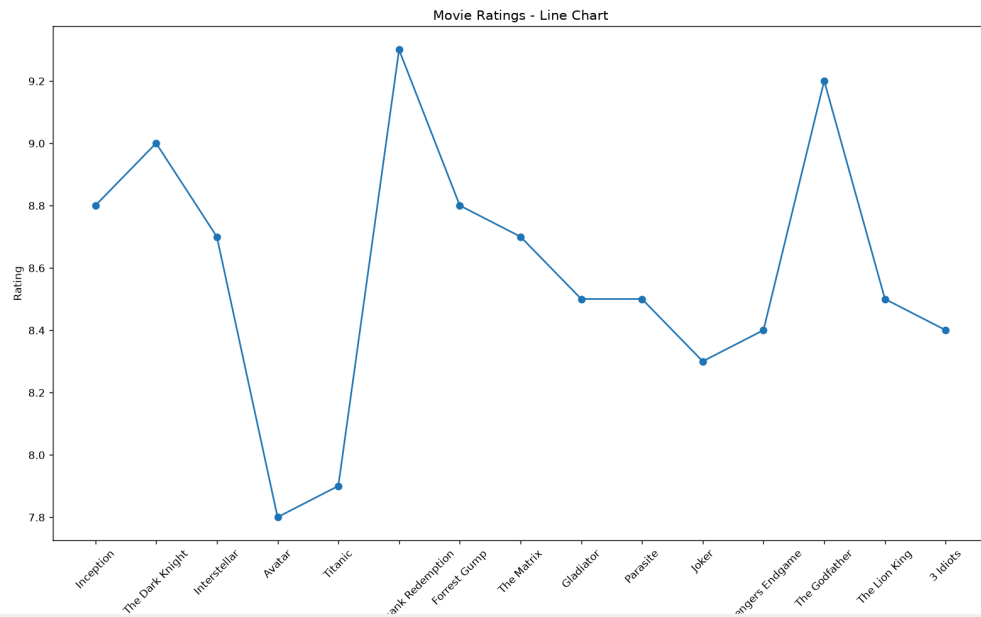
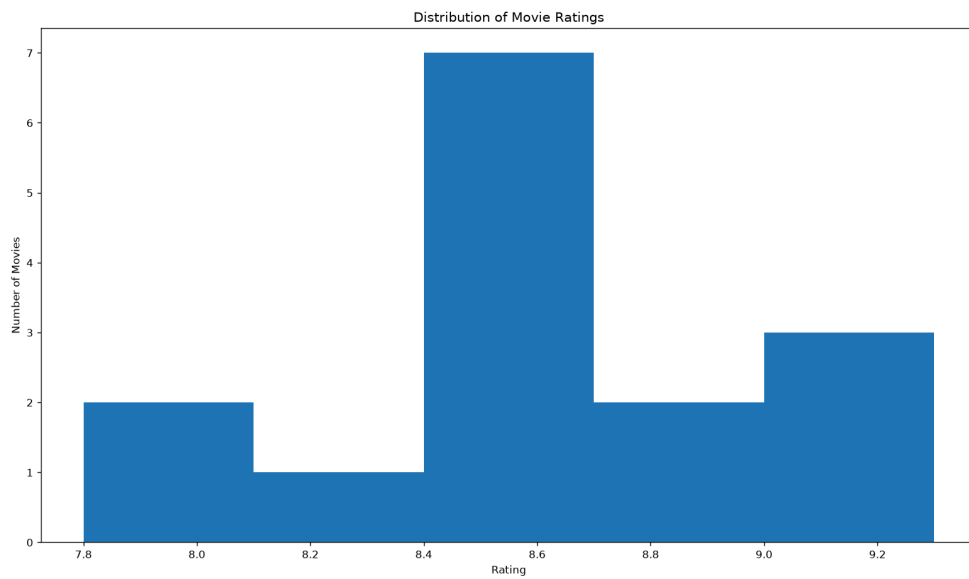


Figure 3



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Figure 4

